

KILKENNY, Ireland

WMO: 039600

Lat: 52.666N

Lon: 7.266W

Elev: 210

StdP: 14.58

Time Zone: 1.00 (IRL)

Period: 98-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) 26.8	(c) 29.0	(d) 24.4	(e) 18.6	(f) 32.1	(g) 26.4	(h) 20.6	(i) 32.5	(j) 30.0	(k) 48.6	(l) 27.5	(m) 49.1	(n) 2.8	(o) 10	(p) 0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.3	(c) 74.7	(d) 63.3	(e) 71.7	(f) 62.0	(g) 69.2	(h) 60.5	(i) 65.4	(j) 71.9	(k) 63.6	(l) 68.9	(m) 62.3	(n) 66.9	(o) 8.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	86.4	67.8	61.6	82.5	66.0	60.2	78.4	64.6	30.2	72.1	29.0	68.9	28.0	66.9	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.9	(b) 21.4	(c) 18.8	(e) 21.2	(f) 80.2	(g) 1.9	(h) 3.6	(i) 19.9	(j) 82.8	(k) 18.8	(l) 85.0	(m) 17.8	(n) 87.0	(o) 16.4	(p) 89.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			20.9	67.8	1.9	1.9	19.5	69.1	18.4	70.2	17.3	71.3	15.9	72.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	42.4	42.3	44.5	47.9	52.6	57.2	60.3	60.3	56.8	51.0	45.4	42.6
	DBStd	8.14	5.20	4.71	4.71	4.73	3.97	3.93	3.90	3.56	4.49	4.91	4.83	5.30
	HDD50	1184	239	219	179	93	19	1	0	0	4	48	150	232
	HDD65	5369	700	637	635	513	384	234	154	151	247	433	587	694
	CDD50	1309	3	2	9	30	101	218	320	321	209	80	13	3
	CDD65	19	0	0	0	0	0	1	9	7	2	0	0	0
	CDH74	120	0	0	0	0	0	10	81	27	2	0	0	0
	CDH80	12	0	0	0	0	0	0	10	2	0	0	0	0
(14) Wind	WSAvg	8.6	10.1	9.6	9.3	9.2	8.8	8.1	7.5	7.0	7.7	8.3	8.5	9.3
(15) Precipitation	PrecAvg	33.1	3.4	2.6	2.4	2.2	2.4	2.0	2.2	2.9	2.9	3.4	3.1	3.5
	PrecMax	44.0	7.9	6.1	5.1	4.6	5.7	4.4	5.5	8.0	7.0	6.3	6.2	7.2
	PrecMin	26.3	0.5	0.1	0.2	0.7	0.5	0.5	0.2	0.1	0.1	0.9	1.1	1.1
	PrecStd	3.9	1.7	1.4	1.2	1.0	1.3	1.1	1.1	1.6	1.6	1.6	1.3	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	56.2	60.1	67.0	71.6	75.7	81.0	77.0	73.4	65.0	58.9	56.4
		MCWB	52.6	52.8	52.1	54.9	59.4	61.5	65.6	66.3	64.3	58.5	55.9	54.0
	2%	DB	53.9	53.8	56.7	62.3	67.3	71.9	76.0	72.9	69.3	62.0	56.7	54.0
		MCWB	51.4	50.8	51.6	53.3	57.5	60.9	64.4	63.2	61.6	57.2	54.2	52.0
	5%	DB	52.5	51.9	54.3	59.3	64.2	69.0	72.2	70.0	66.9	60.2	55.2	52.6
		MCWB	50.2	49.2	49.9	51.4	55.6	59.5	62.8	61.0	60.3	56.6	53.1	50.6
	10%	DB	51.0	50.3	52.5	56.2	61.0	66.1	68.8	67.8	64.8	58.7	53.8	51.2
		MCWB	48.8	47.8	48.7	50.1	54.1	58.1	60.9	60.3	59.1	55.7	51.6	49.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	53.8	54.9	57.1	61.8	64.6	67.6	68.1	65.6	60.3	56.5	54.4
		MCDB	54.8	55.3	58.5	62.7	69.8	71.8	76.4	75.9	70.9	62.1	58.0	56.0
	2%	WB	52.0	51.2	52.5	54.6	59.0	62.1	65.5	65.4	63.4	58.9	54.9	52.5
		MCDB	53.7	53.1	55.1	59.6	65.2	68.9	73.8	69.9	67.5	60.8	56.2	53.8
	5%	WB	50.6	49.7	51.0	52.8	56.7	60.5	63.7	63.3	61.8	57.5	53.3	51.1
		MCDB	52.2	51.5	53.3	57.3	61.7	66.9	69.9	67.3	65.2	59.5	54.8	52.4
	10%	WB	49.1	48.2	49.4	51.1	55.0	59.0	62.0	61.9	60.4	56.0	51.9	49.4
		MCDB	50.8	50.3	52.0	55.6	59.6	64.8	67.3	65.9	63.7	58.2	53.6	50.9
(35) Mean Daily Temperature Range	MDBR		10.7	12.1	13.3	15.1	15.4	15.2	15.3	15.2	14.9	13.0	12.4	10.3
	5% DB	MCDBR	11.0	12.6	14.9	19.8	20.9	20.7	21.8	17.9	16.7	13.7	12.6	11.4
		MCWBR	10.1	10.6	10.3	11.7	12.3	11.3	11.4	9.5	10.3	9.9	10.9	10.6
	5% WB	MCDBR	11.0	11.8	12.6	16.6	17.8	18.2	18.5	15.6	14.7	12.6	11.5	11.4
		MCWBR	10.4	10.7	9.9	10.7	11.3	10.7	10.6	9.4	10.5	10.4	10.7	10.9
(40) Clear-Sky Solar Irradiance	taub		0.333	0.343	0.367	0.377	0.386	0.374	0.380	0.360	0.354	0.344	0.325	0.322
	taud		2.268	2.322	2.280	2.280	2.289	2.351	2.341	2.410	2.426	2.414	2.337	2.237
	Ebn at Noon		201	240	258	271	274	279	274	274	262	238	204	179
	Edh at Noon		21	26	33	37	39	37	37	33	29	24	19	18
(44) All-Sky Solar Radiation	RadAvg		236	454	801	1230	1553	1639	1505	1273	938	539	294	180
	RadStd		23	35	85	119	100	149	135	91	62	48	30	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page